



Matrices (continued) - Exercises Solutions

Exercise 10.1. Calculate the inverses of the following matrices if they exist:

(i) $A = \begin{pmatrix} 1 & 0 \\ 3 & 2 \end{pmatrix}$

Calculating $\det(A)$:

$$ad - bc = (1 \times 2) - (3 \times 0) = 2 - 0 = 2$$

Then

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ -3 & 1 \end{pmatrix}$$

Check:

$$AA^{-1} = \begin{pmatrix} 1 & 0 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -\frac{3}{2} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 1+0 & 0+0 \\ 3-3 & 0+1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

(ii) $B = \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}$

Calculating $\det(A)$:

$$ad - bc = (1 \times 1) - (2 \times -2) = 1 - (-4) = 5$$

Then

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}$$

Check:

$$AA^{-1} = \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{5} & -\frac{2}{5} \\ \frac{2}{5} & \frac{1}{5} \end{pmatrix} = \begin{pmatrix} \frac{1}{5} + \frac{4}{5} & -\frac{2}{5} + \frac{2}{5} \\ -\frac{2}{5} + \frac{2}{5} & \frac{4}{5} + \frac{1}{5} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

(iii) $C = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

Calculating $\det(A)$:

$$ad - bc = (1 \times 4) - (2 \times 3) = 4 - 6 = -2$$

Then

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \frac{1}{-2} \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix}$$

Check:

$$AA^{-1} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix} = \begin{pmatrix} -2+3 & 1-1 \\ -6+6 & 3-2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Exercise 10.2. Let A be the matrix:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

Perform the following row operations on the matrix A (each time starting again with the original matrix).

$$(i) R_1 \leftrightarrow R_2 \quad \begin{pmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \\ 7 & 8 & 9 \end{pmatrix}$$

$$(ii) R_2 \rightarrow 3R_2 \quad \begin{pmatrix} 1 & 2 & 3 \\ 12 & 15 & 18 \\ 7 & 8 & 9 \end{pmatrix}$$

$$(iii) R_3 \rightarrow R_3 + 2R_2 \quad \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 15 & 18 & 21 \end{pmatrix}$$

Exercise 10.3. For each of the matrices below, perform row operations to reduce it to the identity matrix.

$$(a) \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix} \quad (b) \begin{pmatrix} 2 & 3 & 0 \\ 1 & -2 & -1 \\ 2 & 0 & -1 \end{pmatrix} \quad (c) \begin{pmatrix} 1 & 1 & 1 \\ 2 & 3 & 7 \\ 1 & 3 & -2 \end{pmatrix}$$

Below is one possible route to an identity matrix for each example - other methods, that involve applying different operations, will also work.

(a)

$$\begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix}$$

$$R_2 \rightarrow R_2 - (2 \times R_1) \rightarrow \begin{pmatrix} 1 & 3 \\ 0 & -1 \end{pmatrix}$$

$$R_2 \times -1 \rightarrow \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$$

$$R_1 \rightarrow (R_1 - R_2) \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

(b)

$$\begin{pmatrix} 2 & 3 & 0 \\ 1 & -2 & -1 \\ 2 & 0 & -1 \end{pmatrix}$$

$$R_1 \div 2 \rightarrow \begin{pmatrix} 1 & \frac{3}{2} & 0 \\ 1 & -2 & -1 \\ 2 & 0 & -1 \end{pmatrix}$$

$$\begin{array}{l} R_2 \rightarrow (R_2 - R_1) \\ R_3 \rightarrow (R_3 - 2R_1) \end{array} \rightarrow \begin{pmatrix} 1 & \frac{3}{2} & 0 \\ 0 & -\frac{7}{2} & -1 \\ 0 & -3 & -1 \end{pmatrix}$$

$$R_3 \div -3 \rightarrow \begin{pmatrix} 1 & \frac{3}{2} & 0 \\ 0 & -\frac{7}{2} & -1 \\ 0 & 1 & -\frac{1}{3} \end{pmatrix}$$

$$R_2 \leftrightarrow R_3 \rightarrow \begin{pmatrix} 1 & \frac{3}{2} & 0 \\ 0 & 1 & -\frac{1}{3} \\ 0 & -\frac{7}{2} & -1 \end{pmatrix}$$

$$R_3 \rightarrow (R_3 + \frac{7}{2}R_2) \rightarrow \begin{pmatrix} 1 & \frac{3}{2} & 0 \\ 0 & 1 & -\frac{1}{3} \\ 0 & 0 & -\frac{13}{6} \end{pmatrix}$$

$$R_3 \times \frac{6}{13} \rightarrow \begin{pmatrix} 1 & \frac{3}{2} & 0 \\ 0 & 1 & -\frac{1}{3} \\ 0 & 0 & 1 \end{pmatrix}$$

$$R_2 \rightarrow (R_2 + \frac{1}{3}R_3) \rightarrow \begin{pmatrix} 1 & \frac{3}{2} & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$R_1 \rightarrow (R_1 - \frac{3}{2}R_2) \rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

(c)

$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & 3 & 7 \\ 1 & 3 & -2 \end{pmatrix}$$

$$\begin{array}{l} R_2 \rightarrow R_2 - (2 \times R_1) \\ R_3 \rightarrow R_3 - R_1 \end{array} \rightarrow \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 5 \\ 0 & 2 & -3 \end{pmatrix}$$

$$R_3 \rightarrow R_3 - (2 \times R_2) \rightarrow \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 5 \\ 0 & 0 & -13 \end{pmatrix}$$

$$R_3 \div -13 \rightarrow \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{array}{l} R_2 \rightarrow R_2 - (4 \times R_3) \\ R_1 \rightarrow R_1 - R_3 \end{array} \rightarrow \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$R_1 \rightarrow R_1 - R_2 \rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Exercise 10.4. Use Gauss-Jordan elimination to calculate the inverse of the following matrices.

$$(a) \begin{pmatrix} -4 & 2 \\ 3 & 4 \end{pmatrix}$$

$$\left(\begin{array}{cc|cc} -4 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array} \right) \xrightarrow{-\frac{1}{4}R_1} \left(\begin{array}{cc|cc} 1 & -\frac{1}{2} & -\frac{1}{4} & 0 \\ 3 & 4 & 0 & 1 \end{array} \right) \xrightarrow{R_2 - 3R_1}$$

$$\rightarrow \left(\begin{array}{cc|cc} 1 & -\frac{1}{2} & -\frac{1}{4} & 0 \\ 0 & \frac{11}{2} & \frac{3}{4} & 1 \end{array} \right) \xrightarrow{\frac{2}{11}R_2} \left(\begin{array}{cc|cc} 1 & -\frac{1}{2} & -\frac{1}{4} & 0 \\ 0 & 1 & \frac{3}{22} & \frac{2}{11} \end{array} \right) \xrightarrow{R_1 + \frac{1}{2}R_2}$$

$$\rightarrow \left(\begin{array}{cc|cc} 1 & 0 & -\frac{2}{11} & \frac{1}{11} \\ 0 & 1 & \frac{3}{22} & \frac{2}{11} \end{array} \right)$$

$$\text{So } A^{-1} = \begin{pmatrix} -\frac{2}{11} & \frac{1}{11} \\ \frac{3}{22} & \frac{2}{11} \end{pmatrix}.$$

Checking this is correct

$$\begin{pmatrix} -\frac{2}{11} & \frac{1}{11} \\ \frac{3}{22} & \frac{2}{11} \end{pmatrix} \begin{pmatrix} -4 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \checkmark$$

$$(b) \begin{pmatrix} -4 & 2 \\ -1 & -3 \end{pmatrix}$$

$$\begin{aligned} \left(\begin{array}{cc|cc} -4 & 2 & 1 & 0 \\ -1 & -3 & 0 & 1 \end{array} \right) & \xrightarrow{-\frac{1}{4}R_1} \left(\begin{array}{cc|cc} 1 & -\frac{1}{2} & -\frac{1}{4} & 0 \\ -1 & -3 & 0 & 1 \end{array} \right) \xrightarrow{R_2 + R_1} \\ \rightarrow \left(\begin{array}{cc|cc} 1 & -\frac{1}{2} & -\frac{1}{4} & 0 \\ 0 & -\frac{7}{2} & -\frac{1}{4} & 1 \end{array} \right) & \xrightarrow{-\frac{2}{7}R_2} \left(\begin{array}{cc|cc} 1 & -\frac{1}{2} & -\frac{1}{4} & 0 \\ 0 & 1 & \frac{1}{14} & -\frac{2}{7} \end{array} \right) \xrightarrow{R_1 + \frac{1}{2}R_2} \\ \rightarrow \left(\begin{array}{cc|cc} 1 & 0 & -\frac{3}{14} & -\frac{1}{7} \\ 0 & 1 & \frac{1}{14} & -\frac{2}{7} \end{array} \right) \end{aligned}$$

$$\text{So } A^{-1} = \begin{pmatrix} -\frac{3}{14} & -\frac{1}{7} \\ \frac{1}{14} & -\frac{2}{7} \end{pmatrix}.$$

Checking this is correct

$$\begin{pmatrix} -\frac{3}{14} & -\frac{1}{7} \\ \frac{1}{14} & -\frac{2}{7} \end{pmatrix} \begin{pmatrix} -4 & 2 \\ -1 & -3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \checkmark$$

$$(c) \begin{pmatrix} -4 & -4 & -2 \\ 3 & 0 & 4 \\ 1 & 0 & 0 \end{pmatrix}$$

$$\left(\begin{array}{ccc|ccc} -4 & -4 & -2 & 1 & 0 & 0 \\ 3 & 0 & 4 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 \end{array} \right) \quad -\frac{1}{4}R_1$$

$$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 1 & \frac{1}{2} & -\frac{1}{4} & 0 & 0 \\ 3 & 0 & 4 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 \end{array} \right) \quad \begin{array}{l} R_2 - 3R_1 \\ R_3 - R_1 \end{array}$$

$$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 1 & \frac{1}{2} & -\frac{1}{4} & 0 & 0 \\ 0 & -3 & \frac{5}{2} & \frac{3}{4} & 1 & 0 \\ 0 & -1 & -\frac{1}{2} & \frac{1}{4} & 0 & 1 \end{array} \right) \quad -\frac{1}{3}R_2$$

$$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 1 & \frac{1}{2} & -\frac{1}{4} & 0 & 0 \\ 0 & 1 & -\frac{5}{6} & -\frac{1}{4} & -\frac{1}{3} & 0 \\ 0 & -1 & -\frac{1}{2} & \frac{1}{4} & 0 & 1 \end{array} \right) \quad \begin{array}{l} R_1 - R_2 \\ R_3 + R_2 \end{array}$$

$$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & \frac{4}{3} & 0 & \frac{1}{3} & 0 \\ 0 & 1 & -\frac{5}{6} & -\frac{1}{4} & -\frac{1}{3} & 0 \\ 0 & 0 & -\frac{4}{3} & 0 & -\frac{1}{3} & 1 \end{array} \right) \quad -\frac{3}{4}R_3$$

$$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & \frac{4}{3} & 0 & \frac{1}{3} & 0 \\ 0 & 1 & -\frac{5}{6} & -\frac{1}{4} & -\frac{1}{3} & 0 \\ 0 & 0 & 1 & 0 & \frac{1}{4} & -\frac{3}{4} \end{array} \right) \quad \begin{array}{l} R_1 - \frac{4}{3}R_3 \\ R_2 + \frac{5}{6}R_3 \end{array}$$

$$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & -\frac{1}{4} & -\frac{1}{8} & -\frac{5}{8} \\ 0 & 0 & 1 & 0 & \frac{1}{4} & -\frac{3}{4} \end{array} \right)$$

$$\text{So } A^{-1} = \begin{pmatrix} 0 & 0 & 1 \\ -\frac{1}{4} & -\frac{1}{8} & -\frac{5}{8} \\ 0 & \frac{1}{4} & -\frac{3}{4} \end{pmatrix}.$$

Checking this is correct

$$\begin{pmatrix} 0 & 0 & 1 \\ -\frac{1}{4} & -\frac{1}{8} & -\frac{5}{8} \\ 0 & \frac{1}{4} & -\frac{3}{4} \end{pmatrix} \begin{pmatrix} -4 & -4 & -2 \\ 3 & 0 & 4 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \checkmark$$

$$(d) \begin{pmatrix} 4 & 0 & -4 \\ 4 & -1 & 1 \\ 3 & 1 & 2 \end{pmatrix}$$

$$\left(\begin{array}{ccc|ccc} 4 & 0 & -4 & 1 & 0 & 0 \\ 4 & -1 & 1 & 0 & 1 & 0 \\ 3 & 1 & 2 & 0 & 0 & 1 \end{array} \right) \quad \frac{1}{4}R_1$$

$$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & \frac{1}{4} & 0 & 0 \\ 4 & -1 & 1 & 0 & 1 & 0 \\ 3 & 1 & 2 & 0 & 0 & 1 \end{array} \right) \quad \begin{array}{l} R_2 - 4R_1 \\ R_3 - 3R_1 \end{array}$$

$$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & \frac{1}{4} & 0 & 0 \\ 0 & -1 & 5 & -1 & 1 & 0 \\ 0 & 1 & 5 & -\frac{3}{4} & 0 & 1 \end{array} \right) \quad -1R_2$$

$$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & \frac{1}{4} & 0 & 0 \\ 0 & 1 & -5 & 1 & -1 & 0 \\ 0 & 1 & 5 & -\frac{3}{4} & 0 & 1 \end{array} \right) \quad R_3 - R_2$$

$$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & \frac{1}{4} & 0 & 0 \\ 0 & 1 & -5 & 1 & -1 & 0 \\ 0 & 0 & 10 & -\frac{7}{4} & 1 & 1 \end{array} \right) \quad \frac{1}{10}R_3$$

$$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & \frac{1}{4} & 0 & 0 \\ 0 & 1 & -5 & 1 & -1 & 0 \\ 0 & 0 & 1 & -\frac{7}{40} & \frac{1}{10} & \frac{1}{10} \end{array} \right) \quad \begin{array}{l} R_1 + R_3 \\ R_2 + 5R_3 \end{array}$$

$$\rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{3}{40} & \frac{1}{10} & \frac{1}{10} \\ 0 & 1 & 0 & \frac{1}{8} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & -\frac{7}{40} & \frac{1}{10} & \frac{1}{10} \end{array} \right)$$

$$\text{So } A^{-1} = \begin{pmatrix} \frac{3}{40} & \frac{1}{10} & \frac{1}{10} \\ \frac{1}{8} & -\frac{1}{2} & \frac{1}{2} \\ -\frac{7}{40} & \frac{1}{10} & \frac{1}{10} \end{pmatrix}.$$

Checking this is correct

$$\begin{pmatrix} \frac{3}{40} & \frac{1}{10} & \frac{1}{10} \\ \frac{1}{8} & -\frac{1}{2} & \frac{1}{2} \\ -\frac{7}{40} & \frac{1}{10} & \frac{1}{10} \end{pmatrix} \begin{pmatrix} 4 & 0 & -4 \\ 4 & -1 & 1 \\ 3 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \checkmark$$

Exercise 10.5. Write down a system of linear equations corresponding to the situations given.

- (a) Tickets for a play are sold in three areas: standing tickets, seated tickets and balcony tickets. On Thursday night, the venue sold 36 standing tickets, 12 seated tickets and 9 balcony tickets, and made \$612. On Friday, they made \$840 and sold 40 standing tickets, 24 seated tickets and 10 balcony tickets. On Saturday, they sold 2 more of each type of ticket than they did on Friday, and made \$906. How much does each ticket type cost?

Let T_{stand} represent the cost of a standing ticket, T_{seat} represent the cost of a seated ticket, and T_{balc} represent the cost of a balcony ticket. Then we can write an equation for each night:

$$36T_{stand} + 12T_{seat} + 9T_{balc} = 612$$

$$40T_{stand} + 24T_{seat} + 10T_{balc} = 840$$

$$42T_{stand} + 26T_{seat} + 12T_{balc} = 906$$

- (b) At a dog show the dogs were each given a bowl of food, and the audience members and photographers present were given two bowls of food each. Photographers were assigned two seats in the auditorium (so they'd have space for all their kit), audience members 1 seat each, and dogs were asked to share, two to a seat. Altogether, 188 seats were used, 334 bowls of food were eaten, and 535 legs were present (including the legs on each photographer's tripod). All the humans present had two legs. How many dogs, audience members and photographers were there?

Let D represent the number of dogs, A represent the number of audience members, and P represent the number of photographers. Then we can write an equation for each set of items:

$$D + 2A + 2P = 334 \text{ (bowls)}$$

$$0.5D + A + 2P = 188 \text{ (seats)}$$

$$4D + 2A + 5P = 535 \text{ (legs)}$$

- (c) You are playing a quiz game. Getting a question correct gives you 100 points, and getting an answer wrong gives you a penalty of -25 points. You have answered 40 questions, and had a final score of 2250 points. How many questions did you get right and wrong?

Let R represent the number of questions you answered right, and W represent the number of questions you answered wrong. Then we can write these two equations:

$$100R - 25W = 2250$$

$$R + W = 40$$

Exercise 10.6. By finding the inverse of the matrix of coefficients, find solutions to the systems of equations:

$$\begin{array}{lll} (a) \begin{array}{l} -4x_1 + 2x_2 = -22 \\ 3x_1 + 4x_2 = 11 \end{array} & (b) \begin{array}{l} -4x_1 + 2x_2 = 6 \\ -x_1 - 3x_2 = -2 \end{array} & (c) \begin{array}{l} -4x_1 - 4x_2 - 2x_3 = 16 \\ 3x_1 + 4x_3 = -8 \\ x_1 = 0 \end{array} \end{array}$$

(a)

$$\det \begin{pmatrix} -4 & 2 \\ 3 & 4 \end{pmatrix} = -22$$

$$\therefore A^{-1} = -\frac{1}{22} \begin{pmatrix} 4 & -2 \\ -3 & -4 \end{pmatrix} = \begin{pmatrix} -\frac{2}{11} & \frac{1}{11} \\ \frac{3}{22} & \frac{2}{11} \end{pmatrix}$$

therefore

$$\mathbf{x} = \begin{pmatrix} -\frac{2}{11} & \frac{1}{11} \\ \frac{3}{22} & \frac{2}{11} \end{pmatrix} \begin{pmatrix} -22 \\ 11 \end{pmatrix} = \begin{pmatrix} 5 \\ -1 \end{pmatrix}$$

so the solution is $x_1 = 5$ and $x_2 = -1$.

(b)

$$\det \begin{pmatrix} -4 & 2 \\ -1 & -3 \end{pmatrix} = 14,$$

$$\therefore A^{-1} = \frac{1}{14} \begin{pmatrix} -3 & -2 \\ 1 & -4 \end{pmatrix} = \begin{pmatrix} -\frac{3}{14} & -\frac{1}{7} \\ \frac{1}{14} & -\frac{2}{7} \end{pmatrix}$$

therefore

$$\mathbf{x} = \begin{pmatrix} -\frac{3}{14} & -\frac{1}{7} \\ \frac{1}{14} & -\frac{2}{7} \end{pmatrix} \begin{pmatrix} 6 \\ -2 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

so the solution is $x_1 = -1$ and $x_2 = 1$.

(c) The matrix of coefficients is given by

$$A = \begin{pmatrix} -4 & -4 & -2 \\ 3 & 0 & 4 \\ 1 & 0 & 0 \end{pmatrix}$$

Earlier we calculated

$$A^{-1} = \begin{pmatrix} 0 & 0 & 1 \\ -\frac{1}{4} & -\frac{1}{8} & -\frac{5}{8} \\ 0 & \frac{1}{4} & -\frac{3}{4} \end{pmatrix}$$

therefore

$$\mathbf{x} = \begin{pmatrix} 0 & 0 & 1 \\ -\frac{1}{4} & -\frac{1}{8} & -\frac{5}{8} \\ 0 & \frac{1}{4} & -\frac{3}{4} \end{pmatrix} \begin{pmatrix} 16 \\ -8 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ -3 \\ -2 \end{pmatrix}$$

so the solution is $x_1 = 0$, $x_2 = -3$ and $x_3 = -2$.

Exercise 10.7. Using Gauss-Jordan elimination, find solutions to these systems of equations.

(a) $-x_1 + 3x_2 = -2$ $-2x_1 + x_2 = 1$	(b) $3x_1 + x_2 + 2x_3 = 11$ $4x_1 - 4x_3 = -4$ $4x_1 - 2x_2 + x_3 = 13$	(c) $-x_1 - 5x_2 - 2x_3 = -17$ $2x_1 - 2x_2 - 3x_3 = -14$ $3x_1 - x_2 + 4x_3 = -13$
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(a)

$$\begin{aligned}
& \left(\begin{array}{cc|c} -1 & 3 & -2 \\ -2 & 1 & 1 \end{array} \right) \begin{array}{l} -1R_1 \\ \\ \end{array} \longrightarrow \left(\begin{array}{cc|c} 1 & -3 & 2 \\ -2 & 1 & 1 \end{array} \right) \begin{array}{l} \\ R_2 + 2R_1 \\ \end{array} \\
& \longrightarrow \left(\begin{array}{cc|c} 1 & -3 & 2 \\ 0 & -5 & 5 \end{array} \right) \begin{array}{l} \\ -\frac{1}{5}R_2 \\ \end{array} \longrightarrow \left(\begin{array}{cc|c} 1 & -3 & 2 \\ 0 & 1 & -1 \end{array} \right) \begin{array}{l} \\ \\ R_1 + 3R_2 \end{array} \\
& \longrightarrow \left(\begin{array}{cc|c} 1 & 0 & -1 \\ 0 & 1 & -1 \end{array} \right)
\end{aligned}$$

So the solution is $x_1 = -1$ and $x_2 = -1$.

(b)

$$\begin{aligned}
& \left(\begin{array}{ccc|c} 3 & 1 & 2 & 11 \\ 4 & 0 & -4 & -4 \\ 4 & -2 & 1 & 13 \end{array} \right) \begin{array}{l} \frac{1}{3}R_1 \\ \\ \end{array} \longrightarrow \left(\begin{array}{ccc|c} 1 & \frac{1}{3} & \frac{2}{3} & \frac{11}{3} \\ 4 & 0 & -4 & -4 \\ 4 & -2 & 1 & 13 \end{array} \right) \begin{array}{l} \\ R_2 - 4R_1 \\ R_3 - 4R_1 \end{array} \\
& \longrightarrow \left(\begin{array}{ccc|c} 1 & \frac{1}{3} & \frac{2}{3} & \frac{11}{3} \\ 0 & -\frac{4}{3} & -\frac{20}{3} & -\frac{56}{3} \\ 0 & -\frac{10}{3} & -\frac{5}{3} & -\frac{5}{3} \end{array} \right) \begin{array}{l} \\ -\frac{3}{4}R_2 \\ \end{array} \longrightarrow \left(\begin{array}{ccc|c} 1 & \frac{1}{3} & \frac{2}{3} & \frac{11}{3} \\ 0 & 1 & 5 & 14 \\ 0 & -\frac{10}{3} & -\frac{5}{3} & -\frac{5}{3} \end{array} \right) \begin{array}{l} \\ R_1 - \frac{1}{3}R_2 \\ R_3 + \frac{10}{3}R_2 \end{array} \\
& \longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & -1 & -1 \\ 0 & 1 & 5 & 14 \\ 0 & 0 & 15 & 45 \end{array} \right) \begin{array}{l} \\ \\ \frac{1}{15}R_3 \end{array} \longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & -1 & -1 \\ 0 & 1 & 5 & 14 \\ 0 & 0 & 1 & 3 \end{array} \right) \begin{array}{l} R_1 + R_3 \\ R_2 - 5R_3 \\ \end{array} \\
& \longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array} \right)
\end{aligned}$$

So the solution is $x_1 = 2$, $x_2 = -1$ and $x_3 = 3$.

(c)

$$\begin{aligned}
& \left(\begin{array}{ccc|c} -1 & -5 & -2 & -17 \\ 2 & -2 & -3 & -14 \\ 3 & -1 & 4 & -13 \end{array} \right) \xrightarrow{-1R_1} \left(\begin{array}{ccc|c} 1 & 5 & 2 & 17 \\ 2 & -2 & -3 & -14 \\ 3 & -1 & 4 & -13 \end{array} \right) \xrightarrow{\begin{array}{l} R_2 - 2R_1 \\ R_3 - 3R_1 \end{array}} \\
& \rightarrow \left(\begin{array}{ccc|c} 1 & 5 & 2 & 17 \\ 0 & -12 & -7 & -48 \\ 0 & -16 & -2 & -64 \end{array} \right) \xrightarrow{-\frac{1}{12}R_2} \left(\begin{array}{ccc|c} 1 & 5 & 2 & 17 \\ 0 & 1 & \frac{7}{12} & 4 \\ 0 & -16 & -2 & -64 \end{array} \right) \xrightarrow{\begin{array}{l} R_1 - 5R_2 \\ R_3 + 16R_2 \end{array}} \\
& \rightarrow \left(\begin{array}{ccc|c} 1 & 0 & -\frac{11}{12} & -3 \\ 0 & 1 & \frac{7}{12} & 4 \\ 0 & 0 & \frac{22}{3} & 0 \end{array} \right) \xrightarrow{\frac{3}{22}R_3} \left(\begin{array}{ccc|c} 1 & 0 & -\frac{11}{12} & -3 \\ 0 & 1 & \frac{7}{12} & 4 \\ 0 & 0 & 1 & 0 \end{array} \right) \xrightarrow{\begin{array}{l} R_1 + \frac{11}{12}R_3 \\ R_2 - \frac{7}{12}R_3 \end{array}} \\
& \rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 0 \end{array} \right)
\end{aligned}$$

So the solution is $x_1 = -3$, $x_2 = 4$ and $x_3 = 0$.

Exercise 10.8.

(a) Using the dataset $Z = (0, 6, 12, 6)$ calculate the covariance matrix for X , Y and Z where X and Y are the sets given in the previous example. We already know the covariance matrix looks like:

$$\begin{pmatrix} 9 & 7 & \text{cov}(X, Z) \\ 7 & 11.25 & \text{cov}(Y, Z) \\ \text{cov}(Z, X) & \text{cov}(Z, Y) & \text{var}(Z) \end{pmatrix}$$

To calculate the missing entries, we first need to find:

$$\bar{z} = \frac{0 + 6 + 12 + 6}{4} = \frac{24}{4} = 6$$

The variance of Z is given by:

$$s_z^2 = \frac{1}{n} \sum (z - \bar{z})^2 = \frac{(0-6)^2 + (6-6)^2 + (12-6)^2 + (6-6)^2}{4} = \frac{36 + 0 + 36 + 0}{4} = \frac{72}{4} = 18$$

We can also calculate:

$$\begin{aligned}
\text{cov}(X, Z) &= \text{cov}(Z, X) = \frac{\sum (x - \bar{x})(z - \bar{z})}{n} \\
&= \frac{(2-6)(0-6) + (4-6)(6-6) + (6-6)(12-6) + (8-6)(6-6)}{4} \\
&= \frac{24 + 0 + 0 + 0}{4} = \frac{24}{4} = 6
\end{aligned}$$

$$\begin{aligned}
 \text{cov}(Y, Z) &= \text{cov}(Z, Y) = \frac{\sum (y - \bar{y})(z - \bar{z})}{n} \\
 &= \frac{(1 - 6.5)(0 - 6) + (3 - 6.5)(6 - 6) + (5 - 6.5)(12 - 6) + (7 - 6.5)}{4} \\
 &= \frac{33 + 0 + 9 + 0}{4} = \frac{42}{4} = 10.5
 \end{aligned}$$

So the completed covariance matrix is:

$$\begin{pmatrix} 9 & 7 & 8 \\ 7 & 11.25 & 10.5 \\ 8 & 10.5 & 18 \end{pmatrix}$$

(b) A covariance matrix for three variables is given by:

$$\begin{pmatrix} 4 & -2 & 0 \\ -2 & 3 & 1 \\ 0 & 1 & 5 \end{pmatrix}$$

Describe what the matrix tells you about the relationships between the datasets.

This matrix shows that of the three variables X, Y and Z , Z has the largest variance, and Y has the smallest. The covariance of Y and Z is 1 which is positive, so they have a positive correlation. The covariance of X and Y is -2 which is negative, so they have a negative correlation. The covariance of X and Z is zero, so these variables have no correlation.

(c) Calculate the covariance matrix of the datasets $X = (2, 4, 6, 8, 10)$ and $Y = (1, 3, 5, 7, 9)$.

The variance of X is given by:

$$s_x^2 = \frac{1}{n} \sum (x - \bar{x})^2 = \frac{(2 - 6)^2 + (4 - 6)^2 + (6 - 6)^2 + (8 - 6)^2 + (10 - 6)^2}{5} = \frac{16 + 4 + 0 + 4 + 16}{5} = \frac{40}{5} = 8$$

The variance of Y is given by:

$$s_y^2 = \frac{1}{n} \sum (y - \bar{y})^2 = \frac{(1 - 5)^2 + (3 - 5)^2 + (5 - 5)^2 + (7 - 5)^2 + (9 - 5)^2}{5} = \frac{16 + 4 + 0 + 4 + 16}{5} = \frac{40}{5} = 8$$

The covariance of X and Y is given by:

$$\begin{aligned}
 \text{cov}(X, Y) &= \frac{\sum (x - \bar{x})(y - \bar{y})}{n} \\
 &= \frac{(2 - 6)(1 - 5) + (4 - 6)(3 - 5) + (6 - 6)(5 - 5) + (8 - 6)(7 - 5) + (10 - 6)(9 - 5)}{5} \\
 &= \frac{16 + 4 + 0 + 4 + 16}{5} = \frac{40}{5} = 8
 \end{aligned}$$

So the covariance matrix is:

$$\begin{pmatrix} 8 & 8 \\ 8 & 8 \end{pmatrix}$$